Forming Barium Stars through Stable Mass Transfer from TP-AGB Stars

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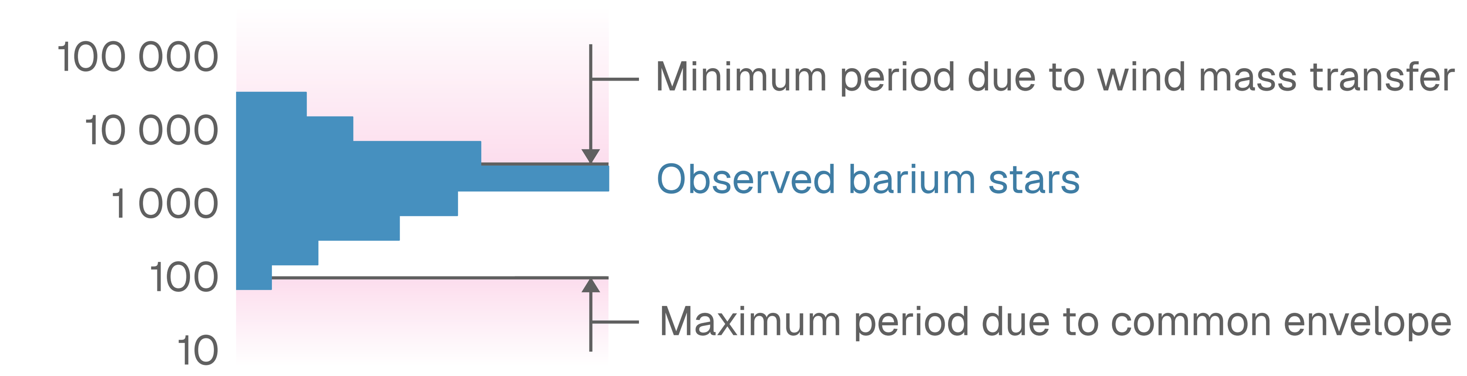
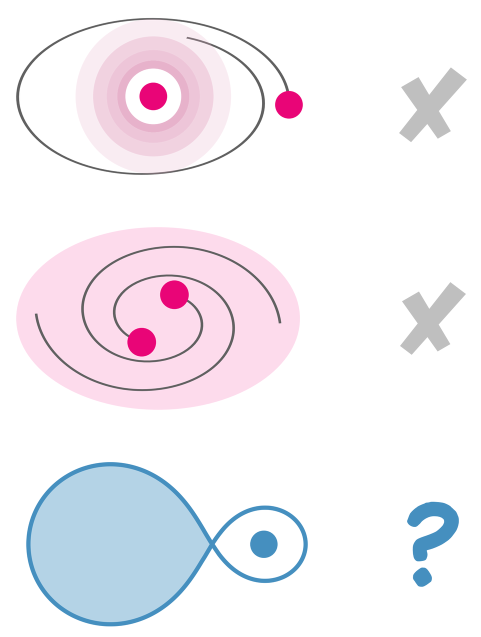
Barium stars are stars enriched in barium by mass accretion from a thermally pulsing (TP) asymptotic giant branch (AGB) star. Barium stars are observed to have periods between 100 and 20 000 days.[1, 2]



The period distribution of barium stars is difficult to explain:



Wind mass transfer alone cannot explain the small orbits.

Common Envelope Events result in orbits that are too small.

Roche lobe overflow (RLOF) mass transfer is often assumed unstable for giants, *but* recent models[3] show it can be stable in RGB & E-AGB stars.

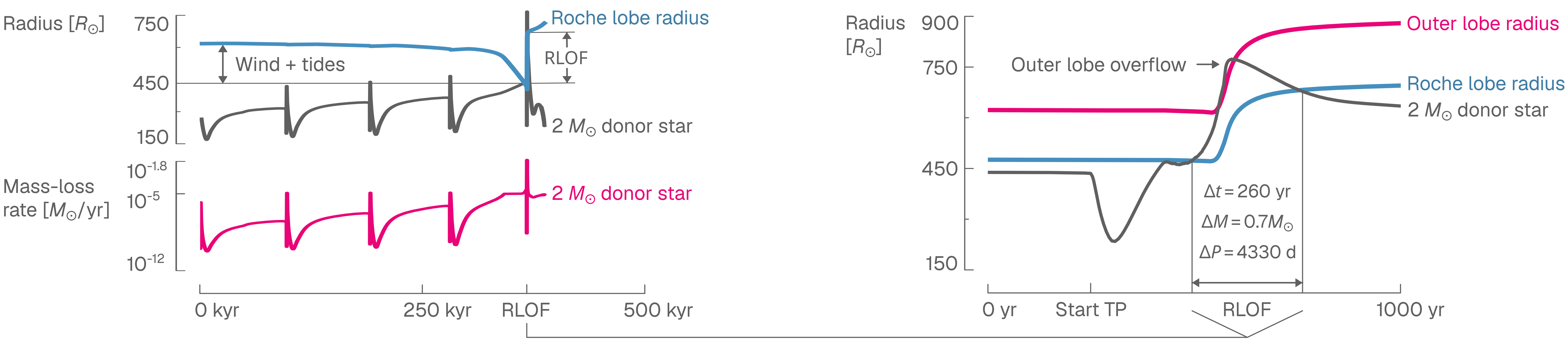
We attempt to undestand how barium stars form, by exploring if TP-AGB RLOF can explain the properties of observed barium stars.

Using *MESA*, we simulate conservative RLOF from a 2 $M_\odot$ TP-AGB star to accretors from 0.8 to 2 $M_\odot$. The models start with periods from 362 to 3986 days and include dredge-up[4], wind mass transfer[5], and tides[6].

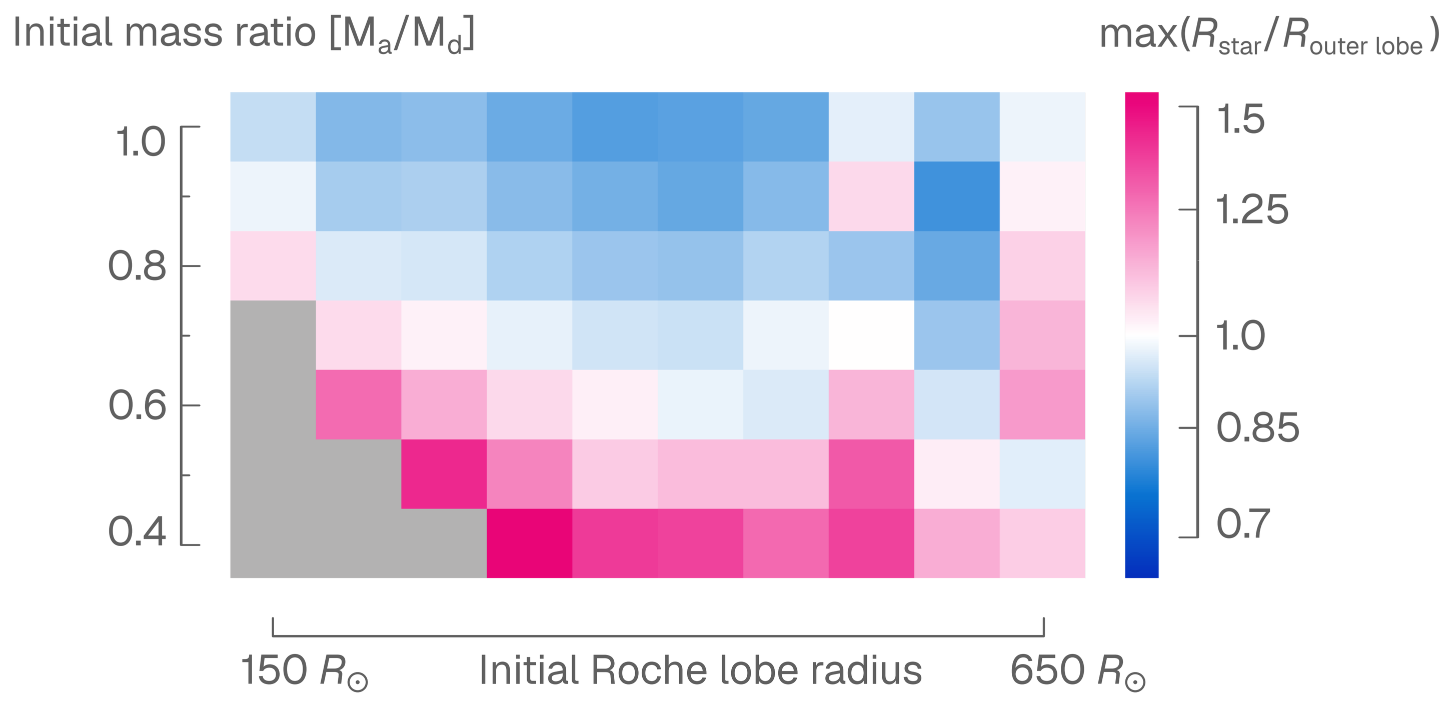
We evaluate the stability of RLOF by looking at the mass-loss rate of the donor, and how much the donor star expands beyond its Roche lobe. We then study how RLOF affects the orbit of the system.

Stellar evolution, together with angular momentum losses due to wind mass loss and tides drive systems into RLOF.

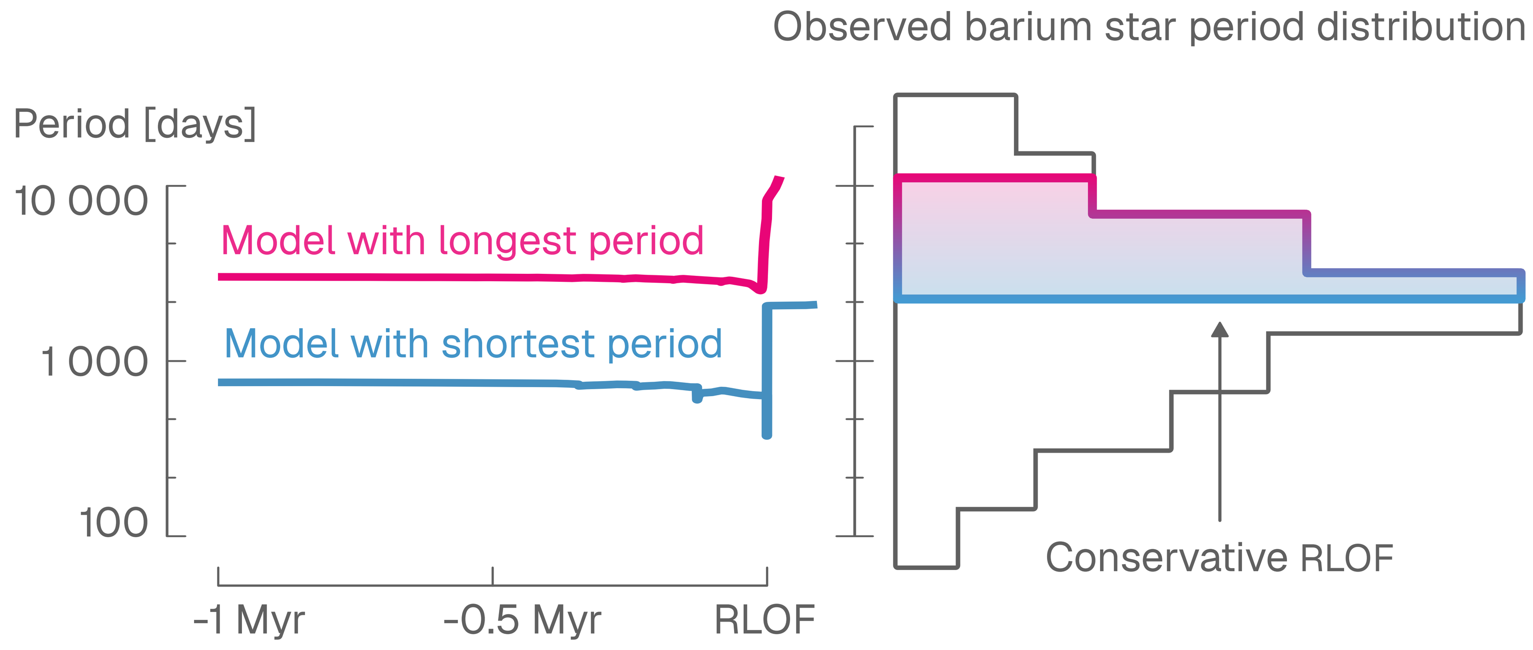
RLOF occurs during the rapid thermal pulses of the TP-AGB star and results in significant mass transfer and widening of the orbit.



A significant portion of our models overflow their outer Roche lobe. *MESA* results are less reliable in this regime.



Fully conservative RLOF mass transfer results in wide orbits, and cannot explain the short periods of many observed barium stars. Non-conservative RLOF is likely required to match observations.



In the future, we want to:

- Explore non-conservative mass transfer and mass loss from outer Lagrange points.
- Investigate mass transfer in systems with non-zero eccentricity.
- Thoroughly compare our results from stable mass transfer to properties of barium stars.

[1]: Jorissen, A., Boffin, H. M. J., Karinkuzhi, D., et al. (2019) A&A626, A127.
[2]: Escorza, A. & De Rosa, R. J. (2023) A&A671, A97.
[3]: Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) A&A669, A45.
[4]: Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) MNRAS527, 9643.
[5]: Saladino, M. I., Pols, O. R., van der Helm, E., Pelupessy, I., & PortegiesZwart, S. (2018) A&A618, A50
[6]: Preece, H. P., Hamers, A. S., Neunteufel, P. G., Schaefer, A. L., & Tout, C. A.(2022) ApJ933, 25.